

Ex.No:03

Wireshark

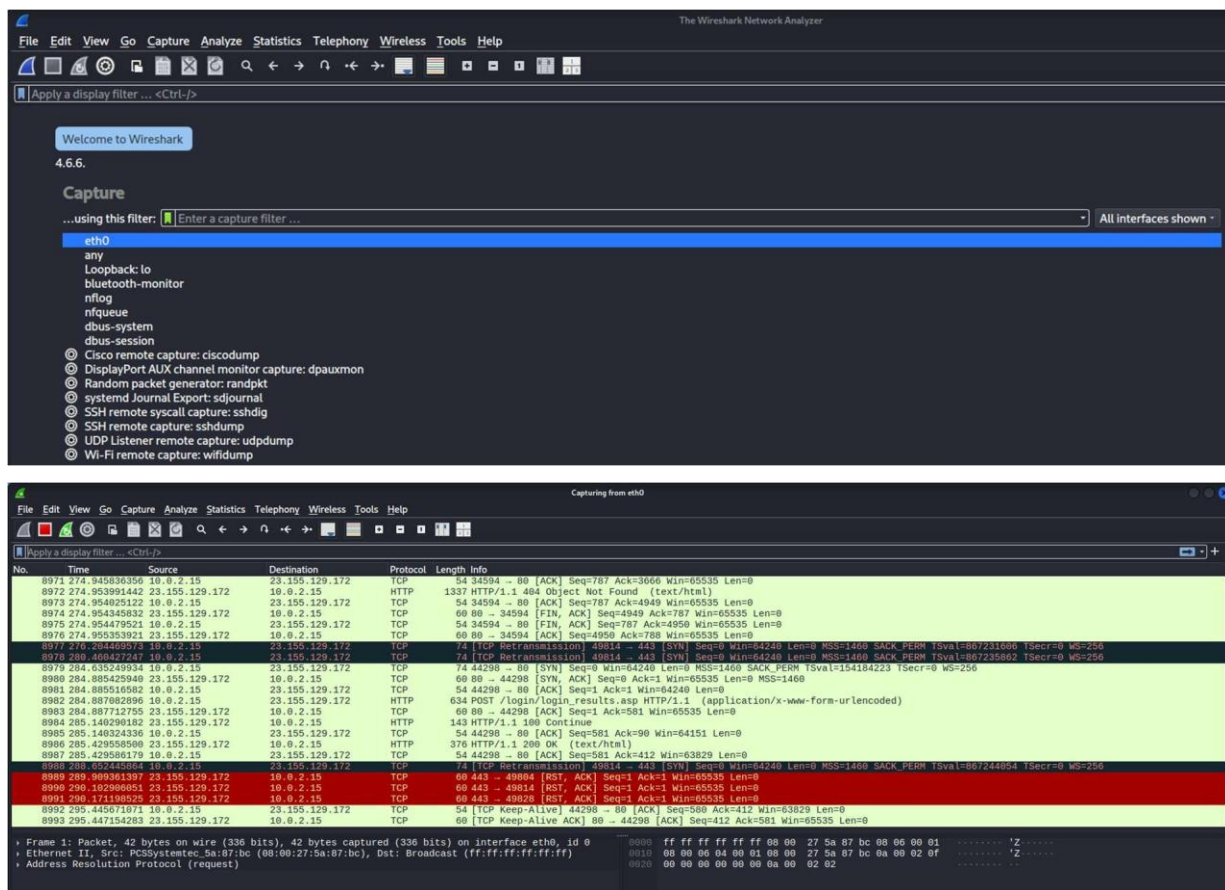
Password Capturing

Wireshark can capture not only passwords but any type of information transmitted over the network: usernames, email addresses, personal information, etc. As long as we can capture network traffic, Wireshark can sniff passing passwords.

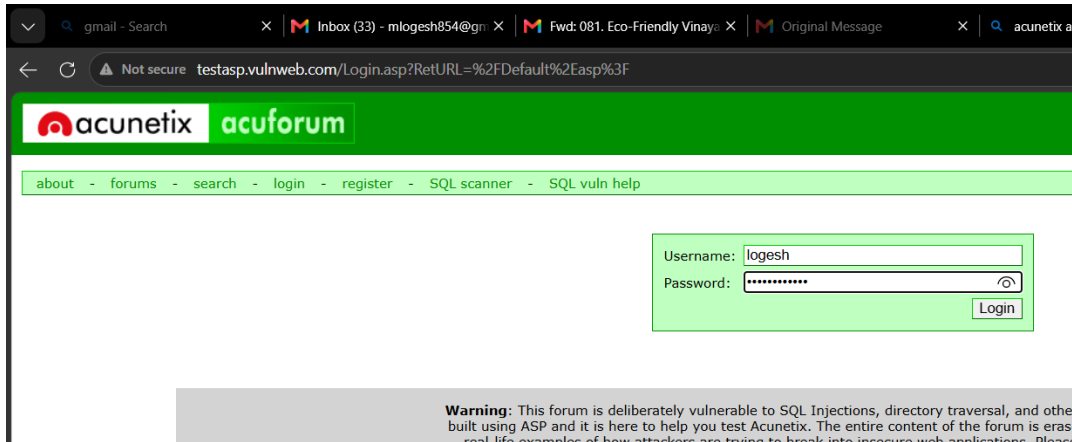
In sniffing can include passwords for various protocols such as HTTP, FTP, Telnet, etc. the captured data can be used to troubleshoot network problems, but can also be used maliciously to gain unauthorized access to sensitive information.

So, here we will see how we can capture the password using the Wireshark network capture analyzer. and see the outputs of the following steps.

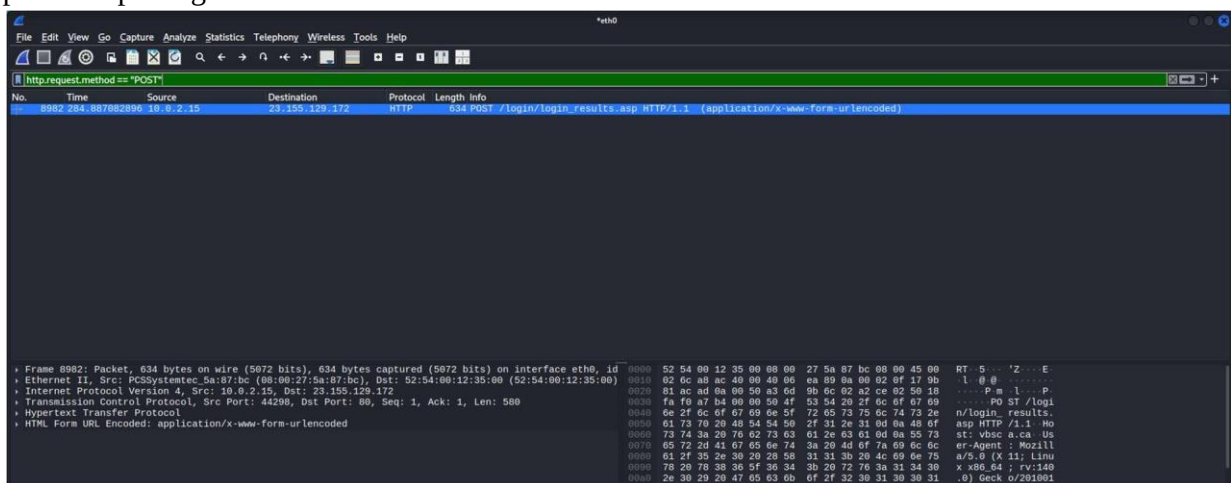
Step 1: First of all, open your Wireshark tool in your window or in Linux virtual machine. and start capturing the network. suppose I am capturing my wireless fidelity.



Step 2: After starting the packet capturing we will go to the website and login the credential on that website as you can see in the image



Step 3: Now after completing the login credential we will go and capture the password in Wireshark. for that we have to use some filter that helps to find the login credential through the packet capturing.



Step 4: Wireshark has captured some packets but we specifically looking for HTTP packets. so in the display filter bar we use some command to find all the captured HTTP packets. as you can http

Step 5: So there are some HTTP packets are captured but we specifically looking for form data that the user submitted to the website. for that, we have a separate filter

As we know that there are main two methods used for submitting form data from web pages like login forms to the server. the methods are-GET

POST

Step 6: So firstly for knowing the credential we use the first method and apply the filter for the GET methods as you can see below.

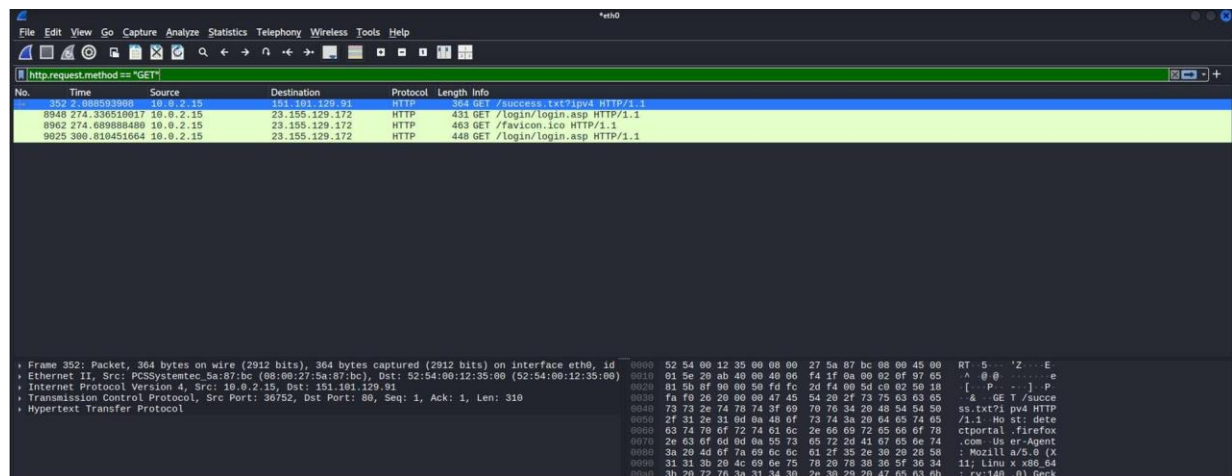
`http.request.method == "POST"` see in the below image the green bar where we apply the filter.

GET method

As you can see in the image there are two packets where the login page was requested with a GET request as well, but there is no form data submitted with a GET request.

Step 7: Now after checking the GET method if we didn't find the form data, then we will try the POST method for that we will apply the filter on Wireshark as you can see.

`http.request.method == "GET"`



As you can see we have a packet with form data click on the packet with user info and the application URL encoded. and click on the down-

HTML form URL Encoded where the login credential is found. login credential as it is the same that we filed on the website in step 2.

Form item: "username" = "mahesh"

Form item: "pass" = "123456"

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Wireshark - Packet 8982 - eth0

> Frame 8982: Packet, 634 bytes on wire (5072 bits), 634 bytes captured (5072 bits) on interface eth0, id 0
> Ethernet II, Src: PCSSystemtec_5a:87:bc (08:00:27:5a:87:bc), Dst: 52:54:00:12:35:00 (52:54:00:12:35:00)
> Internet Protocol Version 4, Src: 10.0.2.15, Dst: 23.155.129.172
> Transmission Control Protocol, Src Port: 44298, Dst Port: 80, Seq: 1, Ack: 1, Len: 580
> Hypertext Transfer Protocol
> HTML Form URL Encoded: application/x-www-form-urlencoded
  > Form item: "txtUsername" = "mahesh"
    Key: txtUsername
    Value: mahesh
  > Form item: "txtPassword" = "123456"

0150  65 6e 74 2d 54 79 70 65 3a 20 61 70 70 6c 69 63  ent-Type : applic
0160  61 74 69 6f 6e 2f 78 2d 77 77 77 2d 66 6f 72 6d  ation/x-www-form
0170  2d 75 72 6c 65 6e 63 6f 64 65 64 0d 0a 43 6f 6e  -urlencoded Con
0180  74 65 6e 74 2d 4c 65 6e 67 74 68 3a 20 33 37 0d  tent-Length: 37
0190  0a 4f 72 69 67 69 6e 3a 20 68 74 74 70 3a 2f 2f  .Origin: http://
01a0  76 62 73 63 61 2e 63 61 0d 0a 43 6f 6e 6e 65 63  vbsca.ca .Connec
01b0  74 69 6f 6e 3a 20 6b 65 65 70 2d 61 6c 69 76 65  tion: keep-alive
01c0  0d 0a 52 65 66 65 72 65 72 3a 20 68 74 74 70 3a  .Referer: http:
01d0  2f 2f 76 62 73 63 61 2e 63 61 2f 6c 6f 67 69 6e  //vbsca.ca/login
01e0  2f 6c 6f 67 69 6e 2e 61 73 70 0d 0a 43 6f 6f 6b  /login.asp .Cook
01f0  69 65 3a 20 41 53 50 53 45 53 53 49 4f 4e 49 44  ie: ASPSESSIONID
0200  53 43 52 51 43 44 54 43 3d 43 4c 48 45 49 46 44  SCRQCDTC =CLHEIFD
0210  41 44 4f 47 45 4e 4b 4b 50 4f 48 47 4a 47 43 41  ADOGENKK POHGJGCA
0220  45 0d 0a 55 70 67 72 61 64 65 2d 49 6e 73 65 63  E. Upgrade-Insec
0230  75 72 65 2d 52 65 71 75 65 73 74 73 3a 20 31 0d  ure-Requests: 1
0240  0a 50 72 69 6f 72 69 74 79 3a 20 75 3d 30 2c 20  .Priority: u=0,
0250  69 0d 0a 0d 0a 74 78 74 55 73 65 72 6e 61 6d 65  i...txt Username
0260  3d 6d 61 68 65 73 68 26 74 78 74 50 61 73 73 77  =mahesh&txtPassw
0270  6f 72 64 3d 31 32 33 34 35 36  ord=1234 56
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